

Kawasaki Supercharged H2 Engine Family

A TECHNICAL REFERENCE FOR CAR INSTALLATIONS

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Please Read First

This document is general guidance only. It is published to help owners, builders and prospective buyers understand the Kawasaki supercharged H2 engine family and what is involved in running one outside its original motorcycle.

It is not a specification, a build manual, or advice on any individual engine or installation. Specifications vary by model year, market and production revision, and engines that have previously been modified or rebuilt may differ from anything described here. Any figure quoted should be verified against the specific engine in front of you before it is relied upon.

Installing a motorcycle engine in a car is a substantial engineering exercise with real consequences if it is done badly. Nothing here replaces proper inspection, measurement and calibration by a competent person.

If you would like advice on a particular engine, car or installation, please get in touch — that conversation is free, and considerably more useful than a general document.

Purpose and Scope

The H2 family is frequently discussed as though it were one engine. It is not.

The variants share a common architecture and a common design philosophy, but the differences between them are more substantial than is generally assumed — and several of them are mechanical rather than electronic. Those differences determine how an engine should be boosted, fuelled and calibrated once the original motorcycle electronics are removed, which in a car installation they always are.

This document is written for car installations generally. The H2 engine has found its way into lightweight sports cars, kit and component cars, sports racing cars, single-seaters, hillclimb specials, buggies and one-off specialist chassis. The packaging, cooling and transmission solutions differ enormously between them. What does not differ is the underlying behaviour of the engine, its lubrication requirements, and its need for a properly specified standalone electrical and management system — and those are the subjects covered here.

Manufacturer figures are Kawasaki-published or press-released values. Where something derives from our own conversion and development work, it is identified as such.

Platform Overview

Kawasaki introduced the supercharged H2 platform in 2015. Every engine in the family is a 998 cc inline-four with a centrifugal supercharger developed in-house by Kawasaki Heavy Industries, drawing on the group's gas turbine and aerospace engineering.

These engines were designed around forced induction from the outset. They are not naturally aspirated engines with a blower added, and the architecture, piston cooling and lubrication strategy all reflect that. It is one of the reasons the platform tolerates car use as well as it does — provided its actual limitations are respected.

Common to all variants	Specification
Configuration	998 cc inline four-cylinder, DOHC, 16-valve
Bore × stroke	76.0 mm × 55.0 mm
Induction	Centrifugal supercharger, in-house Kawasaki design
Supercharger drive	Planetary gear train from the crankshaft via an intermediary gear, stepping impeller speed up to 9.2× crank speed
Impeller	69 mm, forged aluminium, 5-axis CNC machined
Maximum impeller speed	Approaching 130,000 rpm
Charge cooling	None fitted as standard — no intercooler
Piston cooling	Oil jet beneath each piston
Transmission	6-speed, chain final drive
Lubrication (standard)	Wet sump

Kawasaki's position is that the supercharger design does not raise charge temperature sufficiently to require an intercooler on the motorcycle. That reasoning is sound in its original application. **It does not automatically transfer to a car**, where airflow through the engine bay, ambient recirculation, sustained load and heat soak in traffic or on a grid can all be considerably worse than anything a motorcycle sees.

Variant Guide

Model	Role	Compression	Claimed power	Peak torque
Ninja H2 2015–2018	Road-legal hypersport	8.5:1	200 PS (210 PS ram air) @ 11,000 rpm	133.5 Nm
Ninja H2 2019 on	Road-legal hypersport	8.5:1	231 PS (242 PS ram air) @ 11,500 rpm	141.7 Nm
Ninja H2R	Track only	8.3:1	310 PS (326 PS ram air) @ 14,000 rpm	156 Nm @ 12,500 rpm
Ninja H2 SX / SX SE	Sport touring	11.2:1	200 PS (210 PS ram air)	137 Nm @ 9,500 rpm
Z H2 / Z H2 SE	Naked roadster	11.2:1	200 PS @ 11,000 rpm	137 Nm @ 8,500 rpm

The 2019 Ninja H2 uplift came from adopting the H2 SX air filter, intake chamber, spark plugs and ECU. Notably, Kawasaki did not adopt the H2 SX supercharger. The H2 SX engine itself did not receive the 2019 increase and has remained substantially unchanged since its 2018 introduction.

What Actually Differs Between the Variants

This is the section most worth reading carefully, because the common assumption — that these engines differ only in airbox and mapping — is wrong.

Two distinct engine specifications, not one

The family splits into two meaningfully different builds.

Low compression, high boost — Ninja H2 and H2R (8.5:1 / 8.3:1)

Built for peak output. More aggressive camshafts, particularly on the H2R. Higher boost. Greater tolerance of raised boost, because the knock threshold sits further away.

High compression, moderate boost — Ninja H2 SX and Z H2 (11.2:1)

Built for real-world torque and efficiency. These use a different cylinder head, different camshafts, different crankshaft and different pistons from the Ninja H2, along with a different supercharger impeller and plenum chamber. The complete engine is around 3 kg lighter than the Ninja H2 unit.

The superchargers are distinct units between the two groups — not the same part with a different map. Even between the H2 SX and the Z H2 the impeller blade design differs, biasing the Z H2 further towards mid-range torque. On the high-compression engines the impeller carries six full-length blades at the tip with twelve at the base.

Why this matters in a car

Compression ratio is the dominant variable once you are running your own calibration:

- **Boost ceiling.** An 11.2:1 engine reaches its knock limit at markedly less boost than an 8.5:1 engine. If the objective is a high-boost, high-output build, the low-compression Ninja H2 specification is the more tolerant starting point.
- **Fuel sensitivity.** The high-compression engines are more sensitive to fuel quality and to charge temperature — which, as noted above, is a harder problem in a car than on a bike.
- **Off-boost behaviour.** The high-compression engines are considerably better below and between boost. In a car — heavier than a motorcycle, and usually geared longer — this matters far more than it does on the original bike. Torque arriving 1,000 rpm earlier on the Z H2 than the H2 SX is a real driveability advantage in a road or hillclimb car.
- **Calibration is not transferable.** Ignition and fuelling developed on one compression ratio should never be carried across to the other.

There is no single "best" variant. There is a variant that suits the intended use, and choosing it deliberately is worth considerably more than choosing on price or availability.

Z H2, H2 SX and SX SE as donors

As a donor engine, the Ninja H2 SX and SX SE carry the same 11.2:1 specification as the Z H2 — the same high-compression head, camshafts, crankshaft and piston family. The SE suffix is trim, not engine: electronic suspension, quickshifter and lighting. The only mechanical difference between them is the impeller blade design noted above; the intake, exhaust and ECU differences are all discarded in a conversion. Once installed in a car, any of the three is the same engine, and the SX is the more numerous bike. Note that SX models from mid-2022 moved to a Bosch ECU, which is irrelevant with a standalone system but does affect the bike-side tuning market and therefore donor pricing.

Transmission note

The Z H2 uses a dog-ring transmission. Where the original gearbox is retained in a car installation — as it commonly is — this affects shift feel, shift actuation design and expectations about shift quality at low speed. It is worth knowing before the transmission solution is designed around it.

Identifying an Engine Correctly

Do not identify an H2 engine by its engine number prefix.

The ZXT00NE prefix appears across the supercharged family — on Z H2, Ninja H2 SX and Ninja H2 units alike. It identifies the 998 cc supercharged engine family, not the compression ratio or state of tune. Serial numbers run sequentially across the family, so a number falling within a known range proves nothing either.

Given that the two engine groups differ in head, cams, crank, pistons and supercharger, buying on prefix alone is a real risk.

Prefix	Model family
...ZRT00...	Z H2 (ZR1000 series)
...ZXT02...	Ninja H2 SX family
ZX1002... (model code)	Ninja H2, 2019 on

Where an engine has been separated from its motorcycle and no VIN is available, establish the specification physically or by part number before any calibration work begins. Cosmetic finish is not a reliable identifier — supercharger housing colour varies by model and year and is not a Kawasaki specification code.

The gearbox settles it where nothing else is available. First gear is 51/16 on a Ninja H2 or H2R and 40/13 on a Z H2 or Ninja H2 SX, and both are countable with the cassette out.

Lubrication in Car Installations

Why the standard wet sump is the limiting factor

The H2 wet sump is well designed for its intended purpose. A motorcycle leans into a corner, so the oil surface stays broadly perpendicular to the sump and the pickup remains submerged. Sustained lateral loading is simply not a design case.

A car does not lean. The same corner now produces sustained lateral acceleration acting across the sump, and oil migrates away from the pickup and stays there for the duration of the corner. Braking and acceleration add longitudinal displacement on top. The result is intermittent pickup uncovering under precisely the conditions that also produce peak bearing load.

The severity scales with the car. A road-biased build on modest tyres will see far less of it than a light car on slicks sustaining high lateral g through long corners — which is exactly the sort of car this engine tends to end up in. Long-radius corners and banked or high-speed circuits are the worst case, because the loading is sustained rather than transient.

This is a geometry problem, not a capacity problem. It is not reliably solved by baffling, by a deeper pan, or by simply running more oil within the original sump architecture. Oil starvation failures in car-installed H2 engines are a recognised and recurring pattern wherever the original wet sump arrangement has been retained or only lightly modified.

Dry sump conversion — our approach

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Feature	Specification
System type	Custom dry sump
Total system oil volume	Approximately 6 litres
Feed lines	-12 AN
Return / scavenge lines	-12 AN
Scavenge pump	Derived from a proven Suzuki Hayabusa arrangement
Oil pump gears	Up-rated gears (Extreme Creations)
Pressure regulation	External regulation, OEM relief valve retained
Piston oil jets	Cleaned and flow-checked, not modified

The combination of a dry sump, substantially increased system capacity and up-rated pump gearing delivers materially better lubrication security than the standard motorcycle configuration, under conditions the original system was never designed to meet. It also removes the sump depth from the packaging problem, which in a low car is frequently useful in its own right.

Oil Pressure Characteristics

The H2 runs lower oil pressure than several comparable engines — the Suzuki Hayabusa being the usual point of comparison. **This is by design and is not a fault.** The H2 lubrication strategy prioritises flow and piston cooling over headline pressure, and on the high-compression variants the pump is deliberately turned more slowly to reduce parasitic loss.

Condition (hot)	Typical oil pressure
Idle	~2.0 bar
Normal running	~3.0–3.5 bar
Peak rpm	~4.0–4.5 bar

A note on instrumentation. Motorsport pressure sensors frequently indicate apparently low or delayed readings on these engines, because the available sensor location sits downstream in the oil galleries rather than at the pump outlet. Readings should be interpreted against that, not against expectations formed on a different engine family. More than one perfectly healthy H2 has been diagnosed with a problem it did not have.

Engine Speed and Calibration

Factory rev limits vary by variant. The Ninja H2 and H2R run to approximately 14,000 rpm. The Z H2 and H2 SX are limited lower, in the region of 12,000 rpm, consistent with their torque-biased role and higher compression ratio.

In a car installation the OEM loom and ECU are removed entirely, so factory limits are not inherited. Engine speed becomes a deliberate calibration decision rather than an inherited constraint.

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Our own Z H2-based installation is calibrated to 14,000 rpm. The reasoning is that the rotating assembly architecture is common across the family and demonstrably operates at that speed in factory Ninja H2 form — mean piston speed at 14,000 rpm on the 55 mm stroke is approximately 25.7 m/s, a figure Kawasaki already sanctions on this bottom end. The meaningful constraint on the high-compression variants is therefore combustion and thermal loading rather than the mechanical capability of the crank and rods, and that is managed through boost, fuelling and ignition strategy.

This is a documented position on one development engine, arrived at with supporting lubrication and cooling changes already in place. **It should not be read as a general recommendation.** Where an engine is close to standard, on OEM rods, and without the supporting systems to match, the sensible default is to build margin rather than chase the number.

Measured Output and Tuning Headroom

This section answers the question most often asked about the engine: how much power it will make in a car, and which specification to buy for which target. It draws on our own measured output from a car installation and on figures reported consistently by owners and tuners of the motorcycles. Manufacturer claims are treated separately, because they are measured differently.

Measured output from a car installation

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Z H2 engine, 11.2:1, unopened	Measured
At the wheel	245.4 hp @ 13,930 rpm, 105.1 lb-ft @ 10,190 rpm
At the crank	284.4 hp @ 13,830 rpm, 121.9 lb-ft @ 10,190 rpm
Driveline loss	Approximately 13%
Test	Dynojet 224xLC, DIN corrected (1.03), smoothing 5. Daytuner Performance, Harrogate, 21 June 2024

Torque arrives early and stays flat from roughly 8,000 rpm through to beyond 11,000, and power climbs almost linearly to nearly 14,000 rpm without the flattening that normally signals a breathing or fuelling limit. The engine was run with our dry sump system, a standalone MBE ECU with closed-loop lambda and knock control, a car-specification cooling system, its own intake and exhaust, and 99 RON fuel. Both dyno plots are published on the website.

Why manufacturer claims never appear on a dyno

The H2R's 326 PS headline is a ram-air figure — crank output with the intake pressure available at very high road speed. On a dyno the machine is stationary, so that contribution is absent, and 326 PS is 321 bhp before anything else is considered. The factory calibration is also deliberately conservative at the top of the rev range, with rich fuelling and cautious ignition timing, which is entirely correct for an engine sold with a warranty.

The practical result, reported consistently on motorcycle dynos, is that a standard H2R measures 230–240 hp at the rear wheel, and a corrected calibration adds around 30 hp — roughly 300–310 bhp at the crank on a static test. The gap to the brochure is ram air and correction method, not lost performance.

What the H2R actually adds over the H2

The Ninja H2 and H2R share bore, stroke, pistons, valves, valve springs and the supercharger itself. What differs is the intake, the exhaust, the camshafts, the ECU calibration, the head gasket and the clutch. The H2R's higher boost is not a different compressor; it is what the same impeller delivers once the intake and exhaust stop restricting it, combined with a considerably more aggressive ignition map. The H2 airbox and snorkel are the dominant restriction — removing the filter housing alone is reported to be worth around 25 hp on a bike dyno.

This matters in a car because a conversion supplies its own intake, its own exhaust and its own ECU. Most of the H2R's advantage over the H2 is therefore already present in a properly built car installation before the camshafts are considered. The H2R camshafts are the one remaining item. They fit the 8.5:1 engine without further parts — a cam-cover job, not a rebuild —

and are consistently reported to be worth in the region of 20–25 hp at the wheel on an otherwise free-breathing engine, at the cost of some output below 8,000 rpm.

Compression ratio under boost

11.2:1 is an ordinary ratio for a naturally aspirated engine. It is described as high in this family only because, under boost, the static ratio is not the pressure the piston sees. Effective compression scales with absolute intake pressure, and at the same boost the 11.2:1 engine carries more peak cylinder pressure than an H2R.

Engine	Static	Boost	Absolute intake	Effective (approx.)
Ninja H2R	8.3:1	~24 psi	~38.7 psi	~22:1
Ninja H2	8.5:1	~18 psi	~32.7 psi	~19:1
Z H2 / H2 SX	11.2:1	~18 psi	~32.7 psi	~25:1

Kawasaki balances this on the 11.2:1 engines with moderate boost, torque-biased camshafts and a conservative calibration, and there is no intercooler on any variant. Knock is decided by cylinder pressure and charge temperature, not by the static ratio alone. That is why the same engine is comfortable on 99 RON with charge cooling and a proper ignition strategy, and marginal on lower-octane fuel with a fixed map and no cooling. The reported piston ring-land failures on tuned 11.2:1 engines sit almost entirely in the second category: bike ECUs with no ignition-against-boost table, 91–93 AKI fuel, and sustained high output with no charge cooling.

On pistons. It is often stated that the Ninja H2 has forged pistons and the SX and Z H2 have cast. Kawasaki's own engineering description of the H2 records cast pistons throughout, chosen because the alloys with the highest hot strength cannot be forged. The pistons do differ between the two specifications, but the meaningful difference is compression ratio and the thermal margin that follows from it, not the forming process.

The cost argument — stay outside the engine

The reason this engine makes sense in a car is that a used donor is inexpensive. The moment the crankcases are opened — pistons, rods, crank — the cost converges on any other race engine and the advantage is gone. The useful ceiling is therefore the engine as Kawasaki built it, plus the items that stay outside it:

- **Intake, exhaust and ECU** — supplied by the conversion in any case.
- **Camshafts** — a cam-cover job on the 8.5:1 engine.
- **Charge cooling** — the single largest contributor to sustained output and knock margin, and far easier to package in a car than on the motorcycle.
- **Supercharger drive gearing** — a cover-off change, not a case split, giving roughly 5% or 9% more boost depending on stage. This is the step that raises effective compression, and it should only follow charge cooling.

On that basis the measured 284 bhp above, from an unopened 11.2:1 engine, is not a starting point. For a lightweight car it is close to the sensible end point, and it was reached for the price of a crash-damaged motorcycle.

Which engine for which target

Build (car installation, standalone ECU, 14,000 rpm)	At the wheel	At the crank (13% loss)
Z H2 / H2 SX, 11.2:1, unopened — measured	245 hp	284 bhp
Ninja H2, 8.5:1, free intake and exhaust	~255–260 hp	~295–300 bhp
Ninja H2 + H2R camshafts	~265–275 hp	~305–315 bhp
Ninja H2 + camshafts + charge cooling + drive gears	~290–305 hp	~335–350 bhp

Ninja H2 figures are projected from owner and tuner reports on motorcycle dynos, converted at the same 13% driveline factor as our measured run. They are indicative ranges, not measurements.

For the 8.5:1 engine the order matters: camshafts and breathing first, then charge cooling, then boost. Raising boost is the last step, not the first.

The 11.2:1 engine reaches 284 bhp without any of that, and its strength is mid-range on pump fuel rather than peak output. Camshafts with charge cooling are a reasonable further step. Raising boost is the one thing it should not be asked to do, because that is the variable it is already carrying the most of. Whether the H2R camshafts physically fit the 11.2:1 cylinder head has not been verified — the head casting differs from the 8.5:1 engine, and journals, sprockets and chain timing should be checked on the bench before anything is ordered.

Charge cooling and engine management

On these engines the limit on sustained output is thermal, not mechanical. Bike-market intercoolers are compromised by packaging; in a car they need not be. The blower is a centrifugal unit driven at a fixed ratio from the crank, so any pressure lost across the intercooler core is not recovered as it would be on a turbocharged engine with a wastegate — core restriction and pipework matter. Intake air temperature should be measured after the core, because that is the value the ECU's ignition correction should be working from, and it is the one that shows heat soak on a grid or in traffic.

A standalone ECU with closed-loop lambda, an ignition table referenced to manifold pressure and knock control takes most of the calibration risk out of the above. The factory motorcycle ECU has none of these in a form suited to raised output, and that — rather than the engine — accounts for most of the reputation the platform has acquired for fragility when tuned.

Loom and Engine Management

Car installation requires complete removal of the motorcycle wiring loom and ECU. This is not optional, and the reasons are worth understanding before a budget is set:

- **Immobiliser and security integration.** The factory system expects to see motorcycle-specific components before it will permit the engine to run.
- **Ride-by-wire throttle.** Throttle actuation must be correctly reproduced and calibrated by the replacement system.
- **Crank and cam trigger interpretation.** The replacement ECU must be configured for the engine's trigger arrangement.
- **Sensor set and connector standards.** Motorcycle sensors and connectors are not always appropriate to an automotive environment, or to a car's vibration, heat and moisture exposure.
- **Supporting systems.** Cooling fan control, fuel pump control, dash, warning strategy and data logging all require deliberate provision that the bike loom will not give you.

This is where a large proportion of H2 conversion projects stall. A purpose-designed loom and a correctly configured standalone ECU remove most of that difficulty — and they remove it at the start of the project, rather than after the engine is already in the car.

Considerations When Buying an Engine

1. **Establish which variant it is** — by VIN or model code, not engine prefix or appearance. Confirm the compression ratio.
2. **Match the variant to the intended use.** High boost and high output favour the low-compression Ninja H2 specification. Strong mid-range and driveability on pump fuel favour the 11.2:1 Z H2 or H2 SX — and for most lightweight cars the 11.2:1 engine, unopened, is already more than the chassis can use.
3. **Mileage and service history**, particularly oil change discipline.
4. **Evidence of previous car installation.** An engine previously run in a car on a wet sump, or a lightly modified sump, warrants close inspection of bearings and piston oil jets before anything else is spent on it.
5. **Evidence of previous tuning.** An 11.2:1 engine that has been flashed and run hard on the motorcycle warrants a leak-down and bore inspection with the head off, given the ring-land failure pattern described above.
6. **Supercharger condition.** The unit is not user-serviceable and is expensive to replace.
7. **Completeness.** Missing ancillaries are frequently harder to source than the engine itself.
8. **Plan the lubrication and electrical systems before purchase, not after.** Between them they account for most of the cost, most of the difficulty, and nearly all of the failures.

Document Notes

Manufacturer specifications have been checked against Kawasaki-published and press-released figures. Where a statement derives from our own development work rather than manufacturer data, this is identified in the text. Motorcycle dyno figures quoted in the tuning section are drawn from consistent owner and tuner reports and are given as ranges for that reason.

Revision 4, September 2026 — added measured output, tuning headroom, compression under boost, the donor equivalence of Z H2, H2 SX and SX SE, and charge cooling notes. Revision 3, August 2026, was the first public issue.

This document will be revised as the platform and our own development work progress. For advice on engine selection, dry sump conversion, loom and ECU specification, or complete installation support, contact Too Fast To Race.